

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 07 — EXPLORING ICS PACKET CAPTURES

Student Edition

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Lab 07: Exploring ICS Packet Captures

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 4–5 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Use Wireshark display filters to analyze provided ICS packet captures.
- Recognize HTTP, FTP, Modbus, DNP3, EtherNet/IP and S7 communication.
- Identify device, protocol and credential-exposure indicators in recorded traffic.
- Explain security weaknesses visible in legacy or cleartext industrial communications.
- Produce a concise packet-analysis report without interacting with any external system.

Laboratory Environment

System	Expected Address	Account	Password
Analyst/Kali VM	172.16.2.5 (teaching image)	kali	kali

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Step 1. Start the ICDFA analyst VM in an isolated environment. This lab uses instructor-provided packet captures only; no scanning, exploitation or credential reuse is required.

Step 2. Open Wireshark and load the first instructor-provided ICS capture (for the teaching dataset this may be 4SICS-GeekLounge-151021.pcap).

Step 3. Apply the filter http.authbasic. Inspect the HTTP Authorization field and determine whether Basic authentication material is present. If you decode an encoded value for learning purposes, do so only within the provided capture and redact the recovered credential from submitted screenshots.

Step 4. Clear the filter and apply http.request.method. Refine to http.request.method == "POST" and inspect form submissions, request paths and response status codes. Record security-relevant observations without attempting to log in to any identified device.

Step 5. Apply ftp and inspect the FTP session metadata. Record any product/vendor or software-version information visible in the capture.

Step 6. Apply modbus and identify at least two Modbus function codes. If the second teaching capture is supplied, open it and compare the volume/types of Modbus traffic.

Step 7. Apply dnp3 and identify a DNP3 conversation. Record source/destination systems and explain why DNP3 is common in process automation/SCADA environments.

Step 8. Apply enip and inspect available EtherNet/IP/CIP traffic. Record the key header/protocol details visible in the capture.

Step 9. Apply s7comm and inspect an S7 communication exchange. Record source/destination information and any Read Var or comparable operation visible.

Step 10. Complete the Protocol Evidence Matrix for HTTP, FTP, Modbus, DNP3, EtherNet/IP and S7.

Step 11. Write a security assessment identifying at least five observations such as cleartext/encoded credentials, legacy protocols, exposed device metadata, insufficient confidentiality or excessive network visibility.

Step 12. Save your Wireshark workspace/screenshots and submit the report. Do not use any discovered credentials, addresses or device information outside the provided training material.

Required Evidence & Submission

- At least six protocol-analysis screenshots, one for each required protocol family.
- A Protocol Evidence Matrix with filters, endpoints, observed operations and security significance.
- A one-page security assessment prioritizing the observed risks.
- Redacted evidence of any credential material; plaintext credentials must not appear in the submitted report.

Recommended filename: <RegistrationNo>_LAB07_Exploring_ICS_Packet_Captures.pdf

Knowledge Check

1. Why does Base64 encoding not provide confidentiality for HTTP Basic authentication?

Response: _____

2. What information can device banners or protocol metadata reveal to an analyst?

Response: _____

3. How do Modbus, DNP3, EtherNet/IP and S7 differ from ordinary user-facing protocols such as HTTP or FTP?

Response: _____

4. Why is passive packet analysis valuable for understanding an ICS environment?

Response: _____

5. What compensating controls can reduce risk where legacy industrial protocols cannot be replaced immediately?

Response: _____

Marking Rubric — 100 Marks

Criterion	Marks	Standard
Technical execution	50	Lab procedure completed correctly; required configuration/analysis works as intended.
Evidence	20	Screenshots/files are complete, readable, relevant and clearly labelled.
Analysis & knowledge check	20	Responses demonstrate understanding of the industrial-control concepts observed.
Professional reporting	10	Clear structure, correct terminology, concise writing, file naming and submission compliance.

Submission Integrity: Evidence must come from the student's own assigned laboratory environment. Screenshots should show enough context to verify the activity. Do not submit another learner's screenshots or project files.

Solution:



1. Objective

This lab analyses an instructor-provided ICS packet capture (icslabtraffic) using Wireshark display filters, to recognise HTTP, FTP and Modbus communication, identify device/protocol/credential-exposure indicators, and assess the security weaknesses visible in cleartext industrial and IT traffic captured passively, without interacting with any live system.

2. Procedure Summary

The provided capture was opened in Wireshark on the Analyst VM and analysed using the filters http.authbasic, http.request.method, ftp, modbus, dnp3, enip and s7comm in turn. HTTP Basic Authentication material and a cleartext login form submission were identified and screenshotted with credential values redacted. An FTP session revealed both plaintext login credentials and a device banner identifying the FTP server as an AXIS 206 Network Camera. Modbus traffic showed multiple function codes, including a write operation. The dnp3, enip and s7comm filters returned no matching packets, confirming these protocol families are not present in this particular capture; this absence is documented in the Protocol Evidence Matrix below as a genuine finding rather than an incomplete step.

3. Protocol-Analysis Evidence

3.1 HTTP — Basic Authentication

Packet 15 (GET /view/) carries an Authorization: Basic <base64> header, confirming HTTP Basic Authentication material is present and transmitted without encryption.

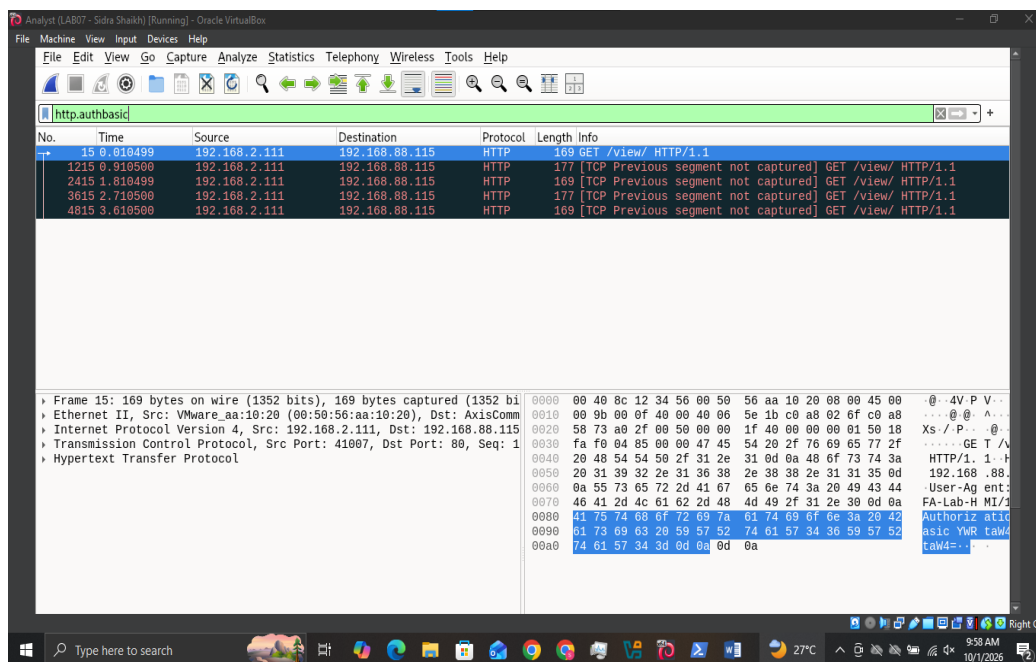


Figure 1. HTTP Basic Authentication header (encoded, not decoded, per redaction policy).

3.2 HTTP — POST request / login form (redacted)

Packet 905 (POST /goform/svLogin) submits an HTML form with Content-Type: application/x-www-form-urlencoded containing username and password fields in cleartext. Field values are redacted in this screenshot; only field names and the fact of exposure are shown, per the lab's credential-redaction requirement.

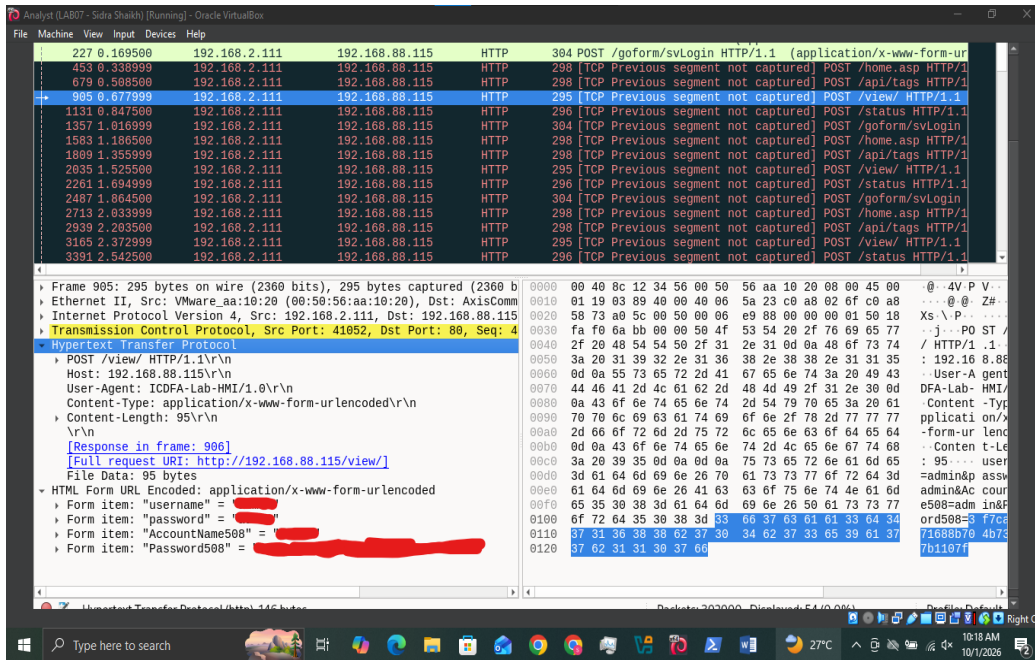


Figure 2. Cleartext login form submission (credential values redacted).

3.3 FTP — session banner and login

The FTP control channel shows a server banner, 220 AXIS 206 Network Camera 4.40 FTP Server ready, disclosing the device's exact make, model and firmware version. USER/PASS login rows are redacted below (credentials were observed in completely unencoded plaintext).

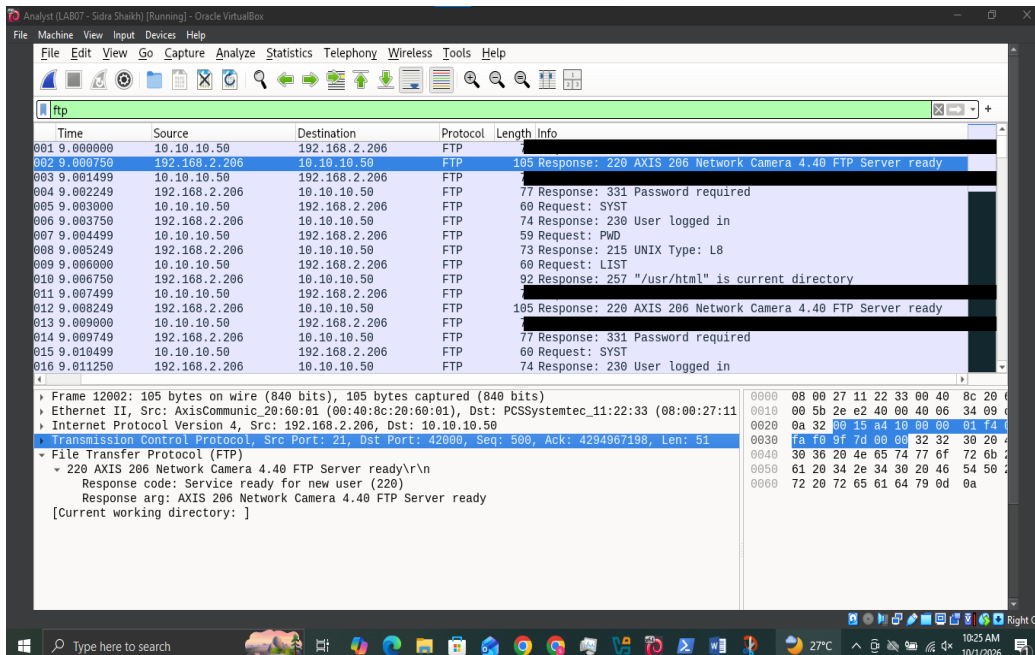


Figure 3. FTP banner disclosing device make/model/firmware (login credential rows redacted).

3.4 Modbus — function codes

Traffic between 192.168.2.111 and 10.10.20.12 shows Function Code 1 (Read Coils, expanded), alongside Function Code 3 (Read Holding Registers), Function Code 6 (Write Single Register) and Function Code 43/1 (Read Device Identification) visible in the surrounding packet list.

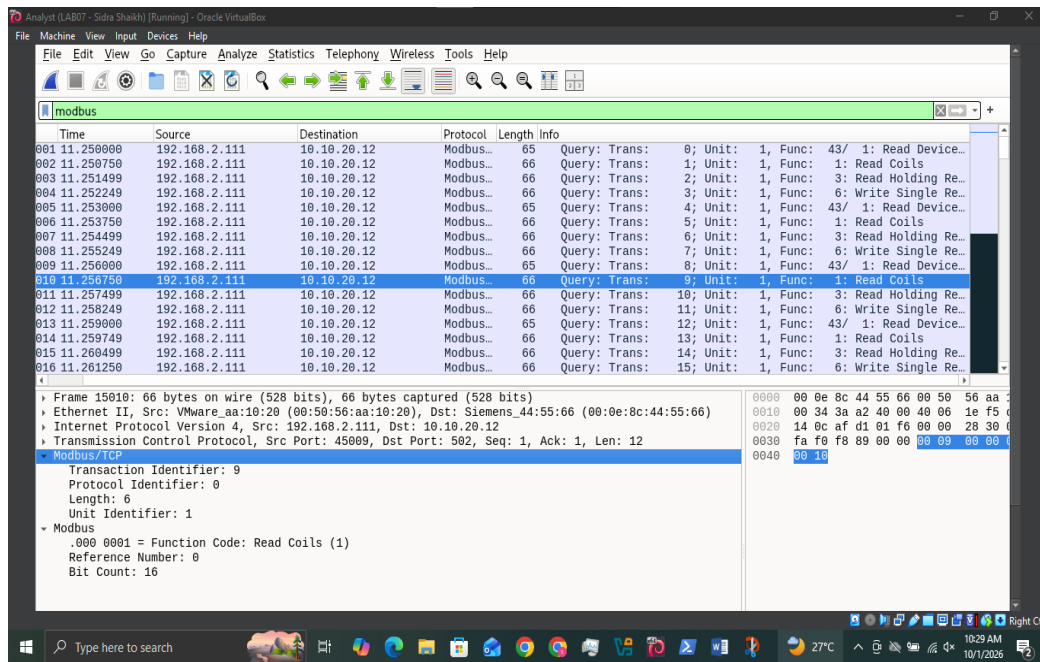


Figure 4. Modbus traffic showing multiple function codes, including a write operation.

3.5 DNP3 — not observed

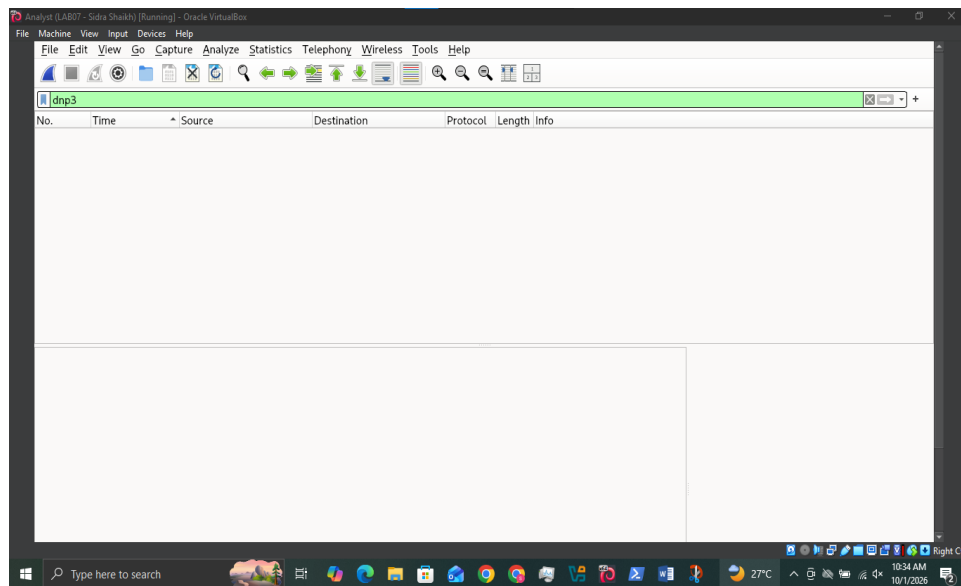


Figure 5. dnp3 filter — no matching packets in this capture.

3.6 EtherNet/IP — not observed

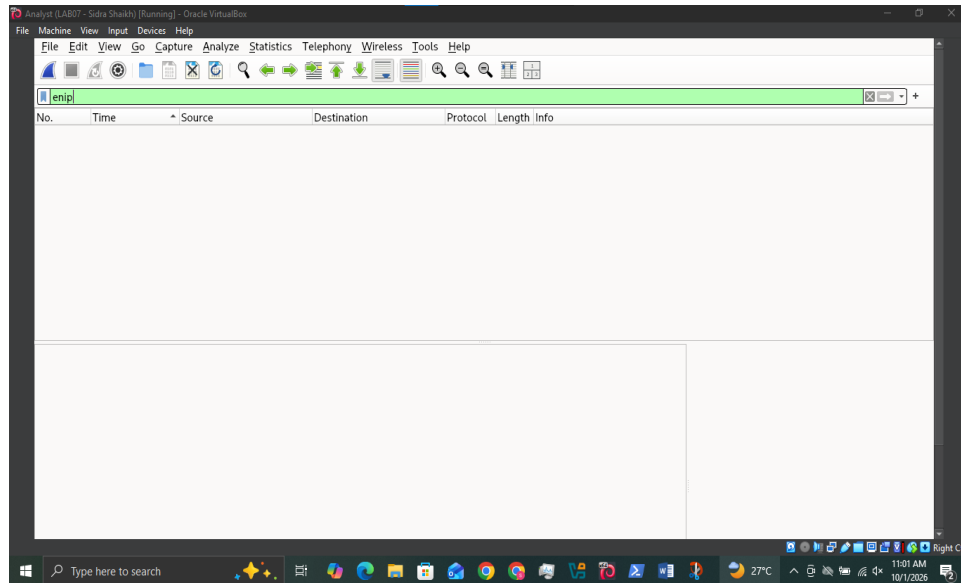


Figure 6. enip filter — no matching packets in this capture.

3.7 S7comm — not observed

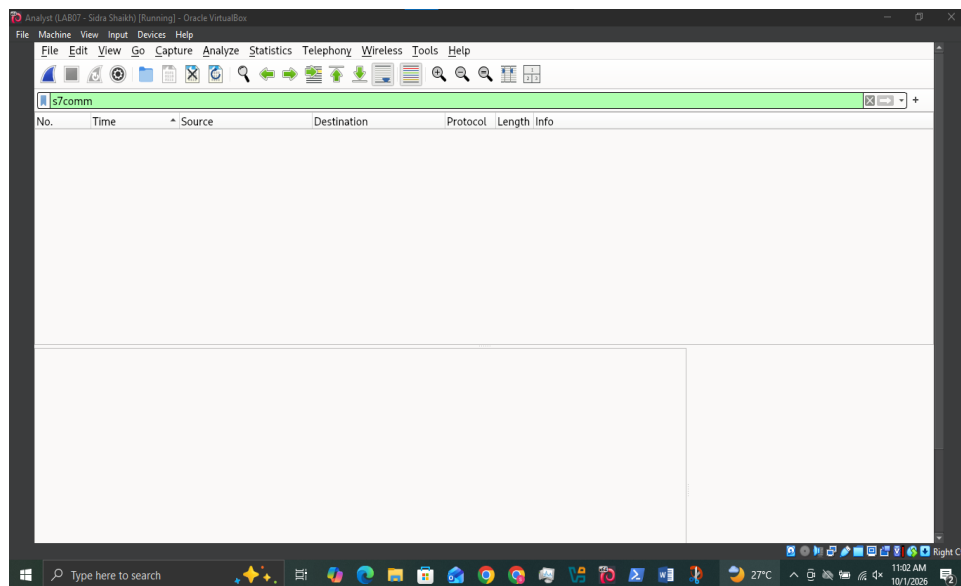


Figure 7. s7comm filter — no matching packets in this capture.

4. Protocol Evidence Matrix

Protocol	Filter	Endpoints	Operation Observed	Security Significance
HTTP	http.authbasic / http.request.method=="POST"	192.168.2.111 ↔ 192.168.88.115	Basic Auth header observed; login form (POST /goform/svLogin) submitted username/password as	Credentials transmitted with no encryption (Base64 is not confidentiality) or in fully unencrypted form data; trivially

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			cleartext form fields.	recoverable by any passive observer.
FTP	ftp	10.10.10.50 ↔ 192.168.2.206	USER/PASS login sequence; server banner identifies device as AXIS 206 Network Camera 4.40 FTP Server.	Credentials sent completely in cleartext (no encoding at all); banner discloses exact make/model/firmware, aiding attacker reconnaissance.
Modbus	modbus	192.168.2.111 ↔ 10.10.20.12	Multiple function codes observed: Read Coils (1), Read Holding Registers (3), Write Single Register (6), Read Device Identification (43/1).	Modbus has no built-in authentication; the presence of a Write function code confirms this traffic could alter process values, and Read Device Identification discloses device metadata, not just process data.
DNP3	dnp3	Not observed	No DNP3 conversations present in the provided capture.	N/A for this capture; DNP3 is typically associated with electric utility / water-system SCADA rather than this facility's apparent HTTP/FTP/Modbus device mix.
EtherNet/IP	enip	Not observed	No EtherNet/IP (CIP) traffic present in the provided capture.	N/A for this capture.
S7comm	s7comm	Not observed	No S7 (Siemens) traffic present in the provided capture.	N/A for this capture.

5. Security Assessment

The following observations are prioritised roughly from most to least severe, based on the credential-exposure and process-control risk each represents.

1. Cleartext login credentials (HTTP form and FTP). Both the web login form (POST /goform/svLogin) and the FTP USER/PASS exchange transmit credentials with no encryption whatsoever. Anyone with passive access to this network segment, as demonstrated by this lab itself, can recover valid login credentials without any active attack.
2. HTTP Basic Authentication is not confidential. The Authorization: Basic header is Base64-encoded, not encrypted; Base64 is trivially reversible and provides no real protection, yet may give operators a false sense of security.
3. Modbus traffic includes an unauthenticated write operation. Function Code 6 (Write Single Register) was observed on the wire. Since Modbus has no built-in authentication, any device with network access, not just the legitimate SCADA system, could in principle issue the same write and alter a live process value.
4. Device identity and firmware are disclosed to any observer. The FTP banner (AXIS 206 Network Camera 4.40) and the Modbus Read Device Identification function code both reveal exact make/model/firmware information. This is reconnaissance-grade information that materially helps an attacker target known vulnerabilities for that specific device and version.
5. Mixed IT and OT traffic on a single accessible segment. HTTP, FTP and Modbus all appear reachable from the same vantage point used for this capture. A flat network that lets a single observer see web-login traffic, file-transfer credentials and live process-control traffic increases the blast radius of any single compromised device or tapped segment.
6. Excessive network visibility. The fact that this entire assessment was possible from a single, passive, unauthenticated capture point, with zero interaction, credentials or prior access, demonstrates insufficient network segmentation and monitoring for the traffic observed.

6. Knowledge Check

1. Why does Base64 encoding not provide confidentiality for HTTP Basic authentication?

Base64 is an encoding scheme, not an encryption algorithm; it has no secret key and is fully and losslessly reversible by anyone using a standard, publicly documented algorithm. Anyone who captures the Authorization: Basic header, as this lab did, can decode it back to the original username:password string in seconds using any Base64 decoder, with no cryptographic work required.

2. What information can device banners or protocol metadata reveal to an analyst?

Banners and metadata can reveal the device's manufacturer, model, firmware/software version, and sometimes its role or directory structure, as seen in this lab's FTP banner (AXIS 206 Network Camera 4.40) and the Modbus Read Device Identification function. For a defender this aids asset inventory; for an attacker the same information narrows down exactly which known vulnerabilities and exploits apply to that specific device and version, making it valuable reconnaissance.

3. How do Modbus, DNP3, EtherNet/IP and S7 differ from ordinary user-facing protocols such as HTTP or FTP?

HTTP and FTP are general-purpose protocols designed for humans and applications to request pages or transfer files, and have at least the option of authentication and, in modern deployments, encryption (HTTPS/FTPS). Modbus, DNP3, EtherNet/IP and S7 are industrial control protocols designed decades ago for closed, trusted serial or local networks, prioritising simplicity, determinism and low latency between a controller and field devices over security; most have no built-in authentication or encryption at all, and were never designed with the assumption that an untrusted party might be on the same network.

4. Why is passive packet analysis valuable for understanding an ICS environment?

Passive analysis lets an analyst build an accurate picture of what devices, protocols and data are actually in use on a network without touching, scanning or risking disruption to fragile or safety-critical ICS equipment, where an active scan could cause a PLC or RTU to behave unpredictably. As this lab demonstrated, a purely passive capture was sufficient to identify device types, firmware versions, credential-exposure issues and even which process-control write operations are possible, all without sending a single packet into the environment.

5. What compensating controls can reduce risk where legacy industrial protocols cannot be replaced immediately?

Where Modbus, FTP or similar cleartext/unauthenticated protocols cannot be upgraded or replaced, risk can be reduced through network segmentation and firewalling (isolating ICS traffic from IT and external networks, as this lab's own VM topology is designed to demonstrate), strict access control to who and what can reach that segment, network monitoring/intrusion detection tuned to these protocols to flag unexpected write commands or logins, VPN or protocol-aware gateways to add authentication and encryption at the network boundary rather than within the legacy protocol itself, and strong physical/administrative controls over who can plug a device into the segment at all.